

EfficientNetB0: A Comparative Study of Modern Robust CNN Architectures for Bean Leaf Disease Detection with Grad-CAM Visualization

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Abstract— Traditional identification of bean leaves' undernourishment by agronomists remains a major challenge due to its variable nature, inaccuracy and ineffectiveness in global bean production. However, recent technological advancement enables rapid detection. The study utilizes the bean leaf dataset containing 2,194 images categorized into 4 classes. Using transfer learning, modern architectures—InceptionV3, ResNet101, DenseNet121, ShuffleNetV2 and EfficientNetB0 were fine-tuned by freezing early layers and training only the final convolutional block alongside a custom classifier, achieving a best-in-class test accuracy of 99.18%, mean f1-score of 0.99 and inference time of 0.48 with the EfficientNetB0 model, followed by 5-fold cross validation of the model to increase reliability and Grad-CAM visualizations for transparent and explainable decisions therefore, illustrating a practical and accessible approach for early and automated detection of nutritional deficiencies in real-world bean farms, facilitating fertilizer optimization, loss minimization, sustainability improvement and long-term protection of this globally significant crop.

Keywords— Bean leaf, Deep Learning, Explainable AI, CNN, Bean Leaf Disease Classification, InceptionV3, ResNet101, DenseNet121, ShuffleNetV2, EfficientNetB0, Grad-CAM

I. INTRODUCTION

Based on recent research, bean crops play a significant role in global agriculture as a major source of protein and essential nutrients. However, bean production is highly affected by various leaf diseases such as blight, rust, mosaic virus, which can significantly reduce crop yield and quality [1]. Early and accurate detection of these diseases is crucial for effective crop management and food security. Moreover, maintaining healthy plants is essential not only for agricultural productivity but also for human survival [1].

Convolutional Neural Network (CNN) approach has automated bean disease identification, demonstrating deep learning feasibility surpassing traditional diagnosis which is slow, costly and limited by individual expertise.

M. Hasan et al. proposed automated disease identification using image-based machine learning techniques on the dataset containing 4,467 images across multiple disease and healthy classes, using smartphone cameras under real field conditions [1]. In another study, a fine-tuned convolutional neural network (CNN) model for

classifying bean leaf diseases using transfer learning techniques was introduced by V. Singh et al. The authors utilize pre-trained models such as MobileNetV2, EfficientNetB6, and NasNet on a dataset containing 1,295 images with three classes of bean diseases. [2]. Moreover, D. W. Girdaw developed a deep convolutional neural network model specifically designed for detecting common bean leaf diseases with pre-trained architectures like AlexNet, GoogleNet, and VGG16 [3]. In another study, Chen et al. developed a standard convolutional neural network for visual-based leaf disease classification [4]. Although these methods demonstrate promising performance, their reliance on single-network architectures or lightweight designs may hinder detailed modeling of complex inter-class variations seen in bean leaf images [2]-[4].

In this study, we implemented CNN models such as InceptionV3, ResNet101, DenseNet121, ShuffleNetV2 and EfficientNetB0 on bean leaf dataset. To enhance robustness and generalization, we preprocessed data with augmentation. The performance of the model is evaluated using metrics like accuracy, precision, recall, F1-score and explainability methods to ensure interpretability.

Research questions:

- Which of the selected CNN architectures demonstrates the highest classification accuracy for bean leaf diseases?
- To what extent can these CNN architectures effectively manage class imbalance and intra-class variability in bean leaf images to ensure robust and reliable disease detection?

II. RELATED WORKS

According to recent research, the use of CNNs with transfer learning has automated the detection of diseases in bean leaves. Hence, reducing the physical examination enhances early diagnosis.

In a study, M. Hasan et al. proposed automated disease identification using image-based machine learning techniques on the dataset containing 4,467 images across multiple disease and healthy classes, using smartphone cameras under real field conditions. However, the study is limited as it does not implement or evaluate any specific

classification model, focusing only on dataset creation, which requires further research for practical deployment [1].

A different study by V. Singh et al. is fine-tuned convolutional neural network (CNN) model for classifying bean leaf diseases using transfer learning techniques. The authors utilize pre-trained models such as MobileNetV2, EfficientNetB6, and NasNet on a dataset containing 1,295 images with three classes of bean diseases. The work evaluates different optimizers and model performances, showing that EfficientNetB6 achieves the highest accuracy of 91.74%, demonstrating the effectiveness of deep learning in agricultural disease detection. However, its limitations include the use of a relatively small and limited dataset, which may reduce generalization capability in real-world scenarios [2].

Moreover, D. W. Girdaw developed a deep convolutional neural network model specifically designed for detecting common bean leaf diseases which consists of multiple convolutional layers and is trained on 1,766 images, focusing on diseases such as brown spot, rust and leaf beetle, comparing the proposed model with pre-trained architectures like AlexNet, GoogleNet, and VGG16, achieving high accuracy (up to 96%), indicating strong performance in disease classification tasks. However, the limitation lies in its focus on a limited number of disease types and controlled dataset conditions, which may not fully represent real-world agricultural variability [3].

Furthermore, Yücel and Yıldırım presented a hybrid feature-fusion model (EfficientNetB0, ResNet101 and ShuffleNet with LDA) that obtained 90.3% accuracy for 885 images classified into 8 classes, where their work faced differences in class and dataset size [5]. Alam et al. [6] achieved promising result using Xception architecture on hog plum leaf disease classification.

Overall, CNN-based and transfer learning methods demonstrated strong performance between 90-96% accuracy. However, they encountered some problems with similarities in symptoms, small datasets, computational requirements and variation in environment, emphasizing the need for larger datasets and more robust, interpretable models.

III. METHODOLOGY

The overall methodological process of our study is shown in Fig. 1. We started with preprocessing the bean leaf disease image dataset. We implemented several modern CNN architectures- InceptionV3, ResNet101, DenseNet121, ShuffleNetV2 and EfficientNetB0 with the same hyperparameter settings. Among all, EfficientNetB0 acquired the highest accuracy of 99.18%. To examine its efficiency, we applied 5-fold cross validation and finally, Grad-CAM visualizations for transparent, explainable decisions.

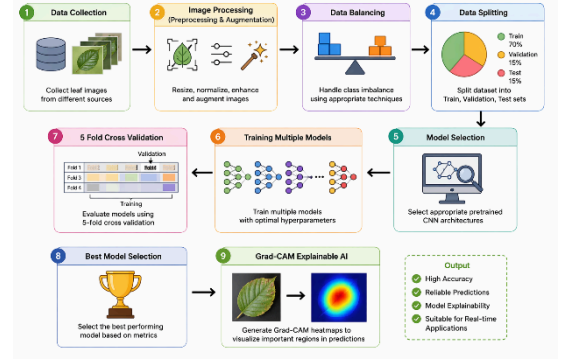


Fig. 1. Methodological framework

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The proposed study utilizes the bean leaf disease dataset, containing 2,194 high-resolution images, captured under various environmental conditions [7], including four distinct classes, Blight (510), Fresh Leaf (554), Mosaic Virus (562) and Rust (568). Fig. 2 shows sample images from the dataset.



Fig. 2. Sample images from bean leaf dataset

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All images were resized to 224×224 pixels to match with the CNN models. Also, we calculated PSNR and SSIM. Before preprocessing, PSNR values were infinite and SSIM values were 1.0, indicating identical images comparison and after preprocessing, PSNR values dropped significantly (21–31 dB) and SSIM values ranged from 0.8 to 0.89,

confirming that preprocessing introduced noticeable pixel-level and structural changes for improving model generalization and robustness. Then, we divided the dataset classes into 70% training, 15% validation and 15% testing. To address the small training subset and improve generalization, data augmentation was applied by generating two augmented copies per training. The training subset was used for updating model weights, while the validation subset was used for tuning hyperparameters and selecting the best model and finally, the test subset for reporting final performance on unseen data. Fig.3 shows sample images after preprocessing.



Fig. 3. Sample images of bean leaves after preprocessing

C. Model Architectures

We evaluated five modern CNN architectures:

- InceptionV3: A CNN, using parallel convolutional filters of different sizes within Inception modules to efficiently capture multi-scale features [8].
- ResNet101: A deep residual network with 101 layers, ResNet uses skip connections (residual blocks) to prevent vanishing gradients, enabling very deep architectures to train effectively. It excels at learning hierarchical representations in images [9].
- DenseNet121: A densely connected CNN where each layer receives feature maps from all preceding layers, improving feature reuse and gradient flow [8].
- ShuffleNetV2: Designed for lightweight and efficient computation, ShuffleNetV2 uses channel shuffling and pointwise group convolutions, making it suitable for mobile or resource-constrained environments while maintaining competitive accuracy [8].
- Proposed Model EfficientNetB0: EfficientNetB0 balances network depth, width and resolution through a compound scaling method, achieving high accuracy with fewer parameters and lower computational cost compared to traditional CNNs [9].

Fig. 4 illustrates the architecture of the proposed model.

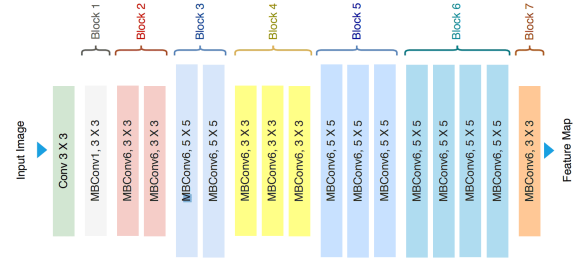


Fig. 4. EfficientNetB0 Model Architecture

D. Training Setup

All models were trained using the cross-entropy loss function and optimized with Adam optimizer (learning rate = $1e-4$, weight decay = $1e-4$). The models were trained for a maximum of 40 epochs with 32 batches and performance was evaluated after each epoch on a validation set. To prevent overfitting, dropout regularization was applied in the classification layers. Additionally, an early stopping was implemented based on validation loss, with a patience of 5 epochs. Thus, the model achieving the lowest validation loss was saved as the best-performing model.

TABLE I. HYPERPARAMETER SETTINGS FOR THE EVALUATED CNN ARCHITECTURES

Parameter	InceptionV3	ResNet101	DenseNet201	ShuffleNetV2	EfficientNetB0
Batch size	32	32	32	32	32
Loss function	Class-balanced focal loss	Class balanced focal loss	Class balanced focal loss	Class balanced focal loss	Class balanced focal loss
Learning rate	1×10^{-4}	1×10^{-4}	1×10^{-4}	1×10^{-4}	1×10^{-4}
Optimizer	AdamW	AdamW	AdamW	AdamW	AdamW
Number of epochs	6	16	13	11	11
Early stopping (patience)	5	5	5	5	5

IV. RESULTS

This section evaluates the performances of the best performing CNN model in the study.

A. EfficientNetB0

Fig. 5 shows the training and validation performance of EfficientNetB0. The loss and accuracy curves show less stable movement, with the validation accuracy of 99.18%.

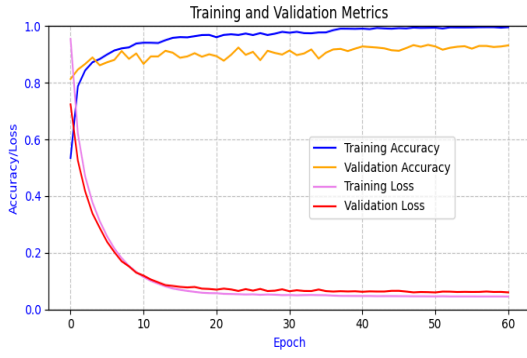


Fig. 5. Accuracy and Loss for training and validation of EfficientNetB0 architecture

In Fig. 6, the confusion matrix implies particularly high accuracy for dominant classes like mosaic virus and rust, the difference might be caused by inherent complexities in the dataset, such as subtle visual patterns.

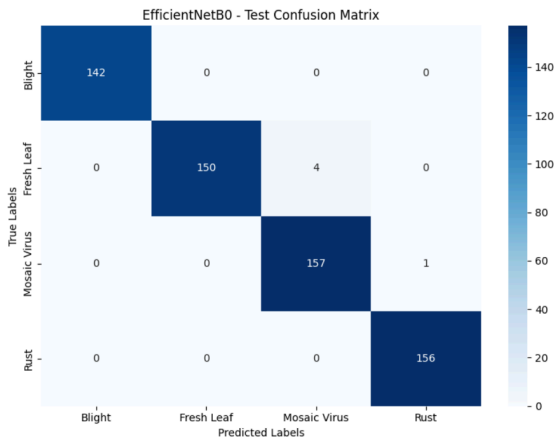


Fig. 6. Confusion Matrix of EfficientNetB0 architecture

EfficientNetB0 demonstrates exceptional classification performance, achieving a mean accuracy of 99.18 with a tight 95% confidence interval of [0.9836, 0.9964], indicating high reliability. Then, model attains near-perfect ROC AUC and PR AUC scores (~ 1.0) across all classes, confirming excellent class separability and precision-recall balance. A Cohen's Kappa of 0.9881 reflects almost perfect agreement, while a low Brier score of 0.0058 indicates well-calibrated predictions. Additionally, the model is computationally efficient, with an average inference time of 0.69 ms per sample, moderate GPU usage (34.4%) and low CPU utilization (10.0%), making it suitable for real-time applications.

V. EXPLAINABLE AI

Ensuring AI models are not only precise and reliable but also interpretable is essential. To accomplish this, we applied Grad-CAM visualization to identify which regions of bean leaf images influence CNN's predictions.

A. Grad-CAM Visualizations

Fig 7. displays original image and Grad-CAM heatmap for the predicted class Mosaic Virus and Blight with confidence: 99.97%. Grad-CAM uses the gradients of the final convolutional layer to identify spatial regions that contribute most to the prediction. Here, red and yellow areas represent regions with the strongest influence on affected areas whereas blue areas indicate low importance on healthy part of leaf.

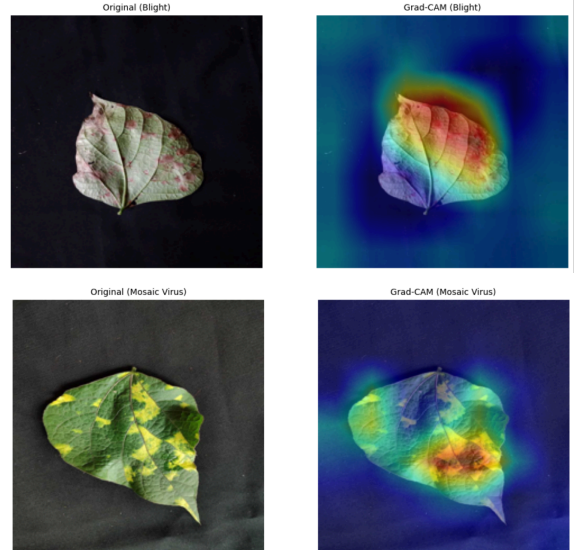


Fig. 7. Grad-CAM Visualization

VI. 5 FOLD CROSS VALIDATION

We applied 5-fold cross validation to further assess the consistency and generalization ability of the proposed EfficientNetB0 architecture [16]. Fig. 8 shows cross-validation accuracy of the best model against each fold. The dataset was segmented into five non-overlapping folds, where each fold was used as test in one iteration while the remaining four were assigned to training. The validation accuracy across five folds stayed uniformly high, ranging roughly between 99.83% and 99.89%, indicating stable training dynamics. The small dispersion across fold results demonstrates that the model is consistent, generalizable and insensitive to data splits.

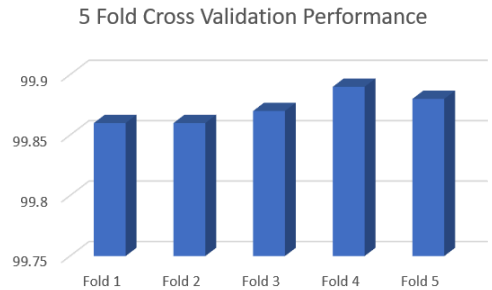


Fig. 8. 5-Fold Cross Validation

VII. COMPARISON AND DISCUSSION

To evaluate performance of our proposed model, we compared the performance of our method against the models used in this work previously.

A. Evaluation of CNN Architecture

We evaluated five CNN architectures— EfficientNetB0 (99.18%), InceptionV3(94.93%), ResNet101 (93.89%), DenseNet121 (97.54%) and ShuffleNetV2 (98.20%) where EfficientNetB0 was seen as the best performing model. However, all models gained high F1 scores for Blight class. Then, finally, all classes' F1 scores improved in EfficientNetB0, achieving the highest macro avg F1 score of 98.89% shows in Table II.

TABLE II. PERFORMANCE COMPARISON OF EVALUATED CNN ARCHITECTURES

Model Name	Classes	Precision	Recall	F1-Score
InceptionV3	Fresh Leaf	88.69	86.19	87.42
	Mosaic Virus	80.56	85.32	88.85
	Blight	85.37	92.11	88.61
	Rust	97.22	87.09	85.37
ResNet101	Fresh Leaf	89.66	84.60	87.05
	Mosaic Virus	88.73	86.62	78.15
	Blight	82.28	85.53	83.87
	Rust	94.32	90.22	92.22
DenseNet121	Fresh Leaf	89.74	83.33	86.42
	Mosaic Virus	82.69	84.31	83.50
	Blight	88.07	95.05	91.43
	Rust	97.92	87.05	86.24
ShuffleNetV2	Fresh Leaf	99.34	98.05	98.69
	Mosaic Virus	94.61	100	97.23
	Blight	99.30	97.90	99.65
	Rust	100	94.87	97.37
EfficientNetB0	Fresh Leaf	100	97.40	98.68
	Mosaic Virus	97.52	99.37	98.43
	Blight	99.98	98.89	99.90
	Rust	99.36	100	99.68

In Fig 9. we can see the comparison between the five architectures, where EfficientNetB0 outperformed all other architectures.

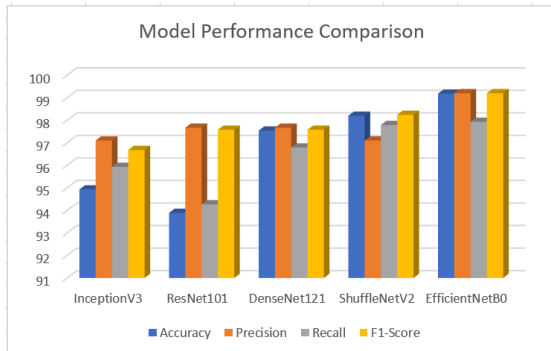


Fig. 9 Comparison between the Architectures

Our CNN model EfficientNetB0 detected bean leaf diseases with an accuracy rate of 99.18%, outperforming several studies that reported accuracies below 98% when benchmarked against recent works. For instance, Ahammed et al. utilized Xception (83.32% accuracy) to identify nutritional deficits in coffee leaves using a dataset of 1,006 images; they applied Grad-CAM for interpretability, but they have noted less effective performance because of the traditional disease models [10]. In another study, Z. Hao et al. developed LeafNet to categorize bean diseases with a 90.16% accuracy [11]. Similarly, X. Wen et al. achieved 91.3% accuracy using a hybrid feature fusion (EfficientNetb0, ResNet101 and ShuffleNet) on 885 images [12]. Finally, after comparing several machine learning and deep learning algorithms, Kamrul *et al.* found ensemble-based machine learning models with the Extra Tree Classifier outperforming CNN's accuracy of 59.08% [13] shows in Table III. Therefore, all these comparisons motivated us to develop a more precise and robust model to achieve higher accuracy in bean disease identification.

TABLE III. COMPARISON OF OUR MODEL WITH THE PREVIOUS WORKS

Author	Method	Accuracy (%)
Ahammed et al. [10]	Xception	83.32
Z. Hao et al. [11]	LeafNet	90.16
X. Wen [12]	EfficientNetB0, ResNet101 and ShuffleNet	90.30
Proposed Models	InceptionV3	94.93
	ResNet101	93.89
	DenseNet121	97.54
	ShuffleNetV2	98.20
	EfficientNetB0	99.18

VIII. CONCLUSION

This research demonstrated the effectiveness of deep learning-based CNN models for automating bean leaf disease detection using provided dataset. As, the dataset initially contained limited number of images; the challenge was addressed through data augmentation techniques which significantly improve model performance and generalization capability. However, among evaluated models, EfficientNetB0 achieved the highest accuracy of 99.18%. Furthermore, 5-fold cross validation was employed to evaluate mode across multiple splits thus, reducing bias and ensuring robustness. Overall, this proposed detection system has strong potential to assist farmers and agronomists in early identification of diseases.

The accuracy of our research work could have been better if there were significant number of pictures in the dataset. Again, with more classes the model could have detected more diseases. However, we focused on the internet available dataset on Bangladesh's bean gardens, ignoring global geographic differences. Moreover, few CNN models were implemented, the model could have been more generalized, if we opted more.

In the future, the research can be further improved by applying more CNN models in sufficient time with owning a global dataset of well-defined images with more classes showing affected area. Again, a bean leaf disease detection application can be implemented for automating and easier detection by using a smartphone. Moreover, ensemble of best performing models can also be utilized to get higher accuracy.

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